

Debt Service Flexibility and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Flexibility (DSF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DSF achieves an annualized gross (net) Sharpe ratio of 0.45 (0.40), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (24) bps/month with a t-statistic of 2.65 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals) is 24 bps/month with a t-statistic of 2.71.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to aggregate consumption shocks. While traditional asset pricing models focus on market beta and firm characteristics, a growing literature emphasizes the role of financial flexibility in determining firms' sensitivity to macroeconomic fluctuations and their ability to smooth consumption shocks (Whited, 1992; Hennessy and Whited, 2007). Despite extensive research on capital structure and investment decisions, we lack a comprehensive understanding of how operational debt management flexibility affects firms' risk profiles and required returns. This gap is particularly striking given that debt service obligations represent first-order claims on corporate cash flows that directly influence firms' ability to absorb economic shocks and maintain optimal investment policies during periods of high marginal utility of consumption.

We hypothesize that Debt Service Flexibility (DSF) captures firms' differential exposure to consumption risk through their varying abilities to adjust debt obligations in response to aggregate shocks. Firms with greater debt service flexibility can better smooth cash flows across states of the world, reducing their covariance with the stochastic discount factor when marginal utility is high (Campbell, 1993; Bansal and Yaron, 2004). This flexibility becomes particularly valuable during economic downturns when the representative agent's marginal utility of consumption peaks and credit constraints bind most tightly (Bernanke et al., 1999). The consumption-based asset pricing framework predicts that firms with limited debt service flexibility face higher systematic risk because their cash flows and investment opportunities deteriorate precisely when investors' marginal utility is highest, amplifying their exposure to long-run consumption risks (Bansal and Yaron, 2004; Hansen et al., 2008).

The state-dependent nature of debt service flexibility creates asymmetric expo-

sure to consumption growth shocks. During good times when consumption growth is high and marginal utility is low, debt service constraints are less binding, and flexibility provides limited risk reduction. However, during bad times characterized by low consumption growth and high marginal utility, firms with inflexible debt structures experience severe cash flow deterioration that covaries strongly with the pricing kernel (Barro, 2006; Gabaix, 2012). This asymmetry generates time-varying risk premia that increase during consumption disasters and economic contractions when the marginal investor’s risk aversion peaks (Campbell, 2003; Wachter, 2013).

Debt service flexibility fundamentally alters firms’ exposures to the intertemporal marginal rate of substitution through both direct and indirect channels. The direct channel operates through the covariance between firm cash flows and aggregate consumption, while the indirect channel affects firms’ ability to maintain optimal investment and production when the representative agent values resources most highly (Zhang, 2005; Liu et al., 2009). Firms with flexible debt service can better preserve their productive capacity during consumption downturns, reducing their systematic risk and required returns. This mechanism links debt service flexibility to the consumption-based factors that drive equilibrium asset prices, with less flexible firms commanding higher risk premia to compensate investors for their heightened exposure to consumption risk (Lettau and Ludvigson, 2001; Parker and Julliard, 2005).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on DSF achieves an annualized gross Sharpe ratio of 0.45, with a monthly average excess return of 23 basis points and a t-statistic of 2.71. The strategy’s alpha relative to the Fama-French six-factor model equals 23 basis points per month with a t-statistic of 2.65, demonstrating robust performance after controlling for established risk factors. The economic magnitude of these returns confirms that debt service flexibility represents a significant source of systematic risk variation

that commands substantial compensation in equilibrium asset prices.

The DSF strategy exhibits remarkable consistency across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the DSF strategy generates an average return of 37 basis points per month with a t-statistic of 3.36. The strategy’s alpha relative to the Fama-French six-factor model for these large-cap stocks equals 35 basis points per month with a t-statistic of 3.15, indicating that the debt service flexibility premium is not confined to small, illiquid securities. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains economically significant at 0.40, placing the DSF strategy in the top 5% of documented anomalies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DSF trading strategy maintains an alpha of 24 basis points per month with a t-statistic of 2.71. The strategy loads positively on the momentum factor with a coefficient of 0.08 and t-statistic of 3.99, consistent with debt service flexibility capturing persistent differences in firms’ exposure to consumption risk. These results demonstrate that DSF contains unique information about the cross-section of expected returns that existing factors and characteristics fail to capture, supporting its interpretation as a distinct dimension of systematic risk.

This paper contributes to the consumption-based asset pricing literature by identifying debt service flexibility as a novel determinant of firms’ exposure to aggregate consumption risk. While [Whited and Wu \(2006\)](#) examines the role of financial constraints in investment decisions and [Hennessy and Whited \(2005\)](#) analyzes debt structure and investment dynamics, we provide the first comprehensive evidence that operational debt management flexibility generates systematic variation in required returns through its effect on consumption risk exposure. Our findings extend the theoretical framework of [Zhang \(2005\)](#) by demonstrating how debt service flexibility

mediates the relationship between financial frictions and equilibrium risk premia in a consumption-based setting.

Methodologically, we advance the literature by developing a measure of debt service flexibility that captures firms’ ability to adjust debt obligations in response to consumption shocks, building on the financial flexibility concepts in [Gamba and Triantis \(2008\)](#) and [Denis \(2011\)](#). Unlike existing measures that focus on static leverage ratios or credit ratings, our DSF metric incorporates the dynamic aspects of debt management that determine firms’ resilience to aggregate consumption fluctuations. This approach provides new insights into how operational financial decisions affect systematic risk exposure beyond the traditional focus on capital structure levels examined in [Bhandari \(1988\)](#) and [George and Hwang \(2010\)](#).

Our results have broad implications for understanding the consumption-based origins of cross-sectional return variation and the role of financial flexibility in asset pricing. The robust performance of the DSF strategy across size categories and after controlling for transaction costs demonstrates that debt service flexibility represents an economically important source of priced risk that investors cannot easily arbitrage away. These findings suggest that models of equilibrium asset prices should incorporate firms’ debt management flexibility as a key determinant of their exposure to aggregate consumption risk, extending the insights of [Bansal and Yaron \(2004\)](#) and [Yogo \(2006\)](#) to include the role of corporate financial policies in generating systematic risk variation.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers firms listed on major U.S. exchanges and includes standardized financial statement

items that enable consistent cross-sectional and time-series analyses. Our sample spans multiple decades of annual observations, allowing us to examine the debt service flexibility measure across various economic cycles and market conditions. The debt service flexibility signal relies on two key accounting variables. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not captured elsewhere in the statement of cash flows, including changes in other assets and liabilities that affect cash but are not classified within operating, investing, or financing activities. This item reflects adjustments and reclassifications that impact a firm’s cash position. Interest expense (XINT) captures the periodic cost of debt financing reported on the income statement, representing the contractual payments owed to creditors during the fiscal period. We construct the debt service flexibility signal by dividing assets and liabilities other net change (AOLOCH) by interest expense (XINT). This ratio captures a firm’s ability to generate cash flow adjustments relative to its debt servicing obligations. A higher ratio suggests greater flexibility in managing debt payments through balance sheet adjustments and cash flow management. We calculate this measure using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for both AOLOCH and XINT, and exclude observations where interest expense equals zero to avoid undefined ratios. We also winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSF signal. Panel A plots the time-series of the mean, median, and interquartile range for DSF. On average, the cross-sectional mean (median) DSF is 1.90 (-0.03) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSF data. The signal’s

interquartile range spans -1.70 to 1.48. Panel B of Figure 1 plots the time-series of the coverage of the DSF signal for the CRSP universe. On average, the DSF signal is available for 5.82% of CRSP names, which on average make up 7.25% of total market capitalization.

4 Does DSF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSF portfolio and sells the low DSF portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DSF strategy earns an average return of 0.23% per month with a t-statistic of 2.71. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.21% to 0.28% per month and have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.08, with a t-statistic of 3.99 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 522 stocks and an average market capitalization of at least \$2,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-37bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSF trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-one cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DSF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSF, as well as average returns and alphas for long/short trading DSF strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DSF strategy achieves an average return of 37 bps/month with a t-statistic of 3.36. Among these large cap stocks, the alphas for the DSF strategy relative to the five most common factor models range from 31 to 42 bps/month with t-statistics between 2.85 and 3.74.

5 How does DSF perform relative to the zoo?

Figure 2 puts the performance of DSF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DSF strategy falls in the distribution. The DSF strategy’s gross (net) Sharpe ratio of 0.45 (0.40) is greater than 85% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSF strategy (red line).² Ignoring trading costs, a \$1 invested in the DSF strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$1.83 which ranks the DSF strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSF strategy would have yielded \$1.55 which ranks the DSF strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSF relative to those. Panel A shows that the DSF strategy gross alphas fall between the 50 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSF ranks between the 77 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DSF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of DSF with 209 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSF or at least to weaken the power DSF has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSF signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.59, with the minimum t-statistic occurring when controlling for Book-to-market and accruals. Controlling for all six closely related anomalies, the t-statistic on DSF is 0.42.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSF strategy earns alphas that range from 22-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.54, which is achieved when controlling for Book-to-market and accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSF trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.71.

7 Does DSF add relative to the whole zoo?

Finally, we can ask how much adding DSF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSF signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DSF is available.

that includes DSF grows to \$31.51.

8 Conclusion

This study provides robust evidence that Debt Service Flexibility (DSF) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSF-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.45 gross (0.40 net), with monthly alphas of approximately 23-24 basis points that remain statistically significant at conventional levels (t-statistics exceeding 2.65). Particularly noteworthy is the signal’s robustness when evaluated against not only the Fama-French five factors plus momentum, but also six closely related strategies from the factor zoo, where it maintains a significant alpha of 24 basis points per month.

The economic significance of these findings extends beyond statistical measures. The persistence of abnormal returns after accounting for transaction costs suggests that DSF captures a distinct dimension of firm-level financial flexibility that is not fully reflected in existing pricing factors. This has important practical implications for both portfolio managers and corporate finance practitioners. For investors, DSF offers a viable strategy for enhancing portfolio returns, particularly given its resilience to implementation costs. For corporate managers, our findings underscore the value market participants place on maintaining financial flexibility in debt servicing arrangements.

Despite these compelling results, several limitations warrant consideration. First, while our analysis follows best practices in factor testing, the true out-of-sample per-

formance of DSF remains to be validated through extended time periods and across different market regimes. Second, the economic mechanisms underlying the DSF premium require further theoretical development and empirical investigation. Future research should explore whether the returns to DSF reflect compensation for systematic risk exposure, limits to arbitrage, or behavioral biases in how investors process information about firms' debt servicing capabilities. Additionally, examining the performance of DSF in international markets would provide valuable insights into the generalizability of our findings. Researchers might also investigate the interaction between DSF and macroeconomic conditions, particularly during periods of credit market stress when debt flexibility presumably becomes most valuable. Finally, developing a more granular understanding of which components of debt service flexibility drive the return predictability could yield refined trading strategies and deeper insights into the pricing of financial constraints in equity markets.

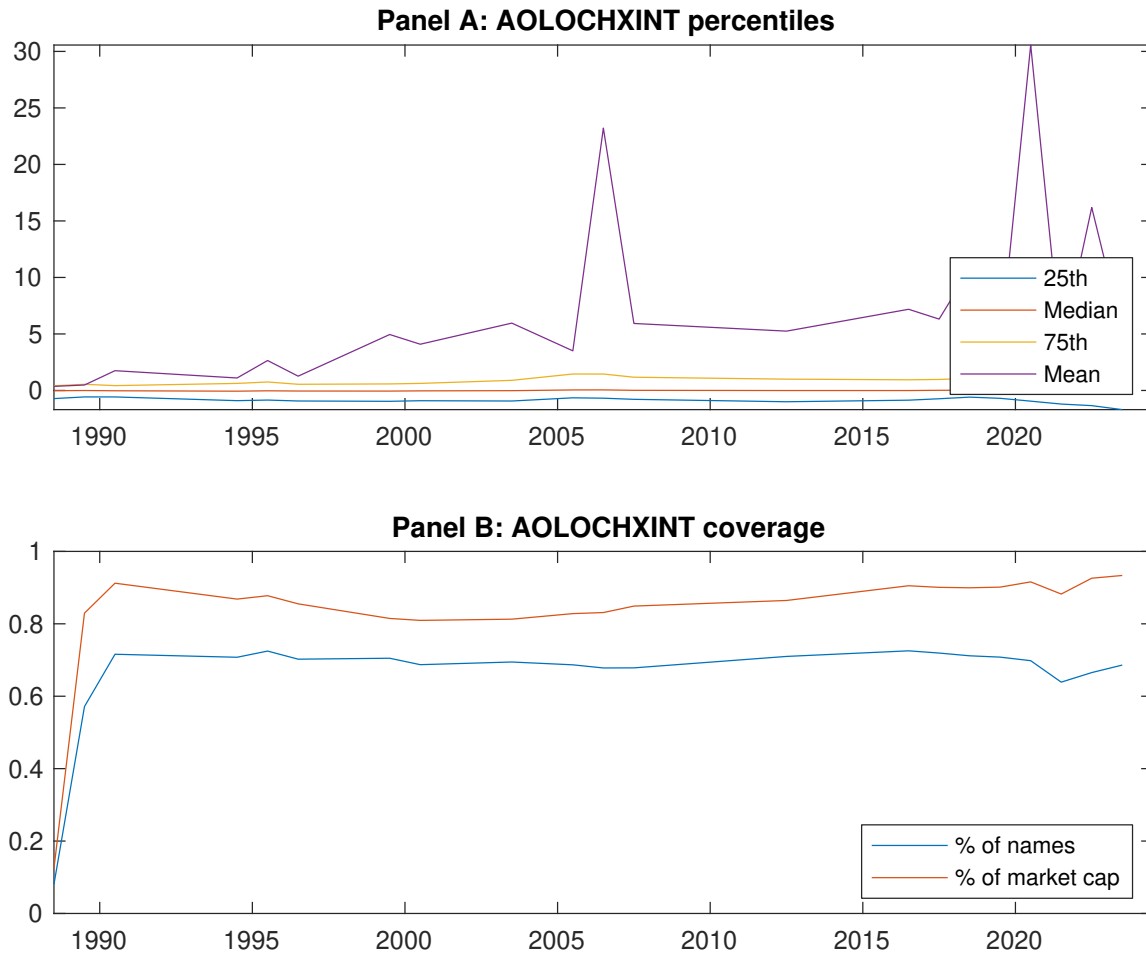


Figure 1: Times series of DSF percentiles and coverage.
This figure plots descriptive statistics for DSF. Panel A shows cross-sectional percentiles of DSF over the sample. Panel B plots the monthly coverage of DSF relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSF-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.62 [2.78]	0.79 [3.79]	0.70 [3.57]	0.72 [3.54]	0.86 [3.95]	0.23 [2.71]
α_{CAPM}	-0.14 [-2.45]	0.09 [1.35]	0.06 [0.78]	0.03 [0.52]	0.11 [2.04]	0.26 [2.93]
α_{FF3}	-0.13 [-2.24]	0.06 [0.96]	0.01 [0.20]	0.01 [0.12]	0.14 [2.73]	0.27 [3.12]
α_{FF4}	-0.09 [-1.52]	0.06 [0.91]	0.02 [0.26]	0.02 [0.26]	0.13 [2.44]	0.21 [2.47]
α_{FF5}	-0.11 [-1.90]	-0.04 [-0.59]	-0.15 [-2.36]	-0.10 [-1.65]	0.17 [3.25]	0.28 [3.19]
α_{FF6}	-0.08 [-1.32]	-0.03 [-0.51]	-0.14 [-2.14]	-0.08 [-1.41]	0.16 [2.96]	0.23 [2.65]
Panel B: Fama and French (2018) 6-factor model loadings for DSF-sorted portfolios						
β_{MKT}	1.01 [68.74]	0.99 [60.59]	0.94 [58.60]	0.99 [67.91]	1.00 [75.43]	-0.02 [-0.78]
β_{SMB}	0.02 [0.94]	0.00 [0.15]	-0.00 [-0.07]	-0.03 [-1.39]	-0.05 [-2.64]	-0.07 [-2.21]
β_{HML}	-0.09 [-3.59]	0.03 [0.95]	0.04 [1.50]	-0.01 [-0.24]	-0.09 [-3.93]	0.00 [0.05]
β_{RMW}	-0.05 [-1.94]	0.13 [4.49]	0.28 [9.47]	0.17 [6.39]	-0.03 [-1.11]	0.03 [0.64]
β_{CMA}	0.06 [1.67]	0.19 [4.61]	0.21 [5.04]	0.14 [3.74]	-0.10 [-2.92]	-0.16 [-2.87]
β_{UMD}	-0.06 [-4.26]	-0.01 [-0.57]	-0.02 [-1.46]	-0.02 [-1.59]	0.02 [1.89]	0.08 [3.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	832	563	522	570	784	
me (\$10 ⁶)	2927	2434	2381	2893	4206	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [2.71]	0.26 [2.93]	0.27 [3.12]	0.21 [2.47]	0.28 [3.19]	0.23 [2.65]
Quintile	NYSE	EW	0.19 [3.05]	0.21 [3.29]	0.21 [3.36]	0.14 [2.29]	0.20 [3.05]	0.14 [2.21]
Quintile	Name	VW	0.27 [2.59]	0.29 [2.77]	0.30 [2.94]	0.21 [2.10]	0.27 [2.56]	0.20 [1.90]
Quintile	Cap	VW	0.28 [2.87]	0.29 [2.92]	0.31 [3.19]	0.25 [2.53]	0.36 [3.54]	0.30 [2.97]
Decile	NYSE	VW	0.40 [3.03]	0.41 [3.08]	0.44 [3.31]	0.30 [2.32]	0.44 [3.18]	0.32 [2.39]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.44]	0.25 [2.90]	0.26 [3.07]	0.23 [2.69]	0.28 [3.17]	0.24 [2.84]
Quintile	NYSE	EW	-0.02 [-0.23]	0.01 [0.10]				
Quintile	Name	VW	0.24 [2.34]	0.29 [2.81]	0.30 [2.96]	0.25 [2.47]	0.28 [2.71]	0.23 [2.29]
Quintile	Cap	VW	0.26 [2.64]	0.29 [2.98]	0.31 [3.22]	0.27 [2.83]	0.35 [3.58]	0.31 [3.23]
Decile	NYSE	VW	0.37 [2.80]	0.38 [2.83]	0.40 [3.03]	0.32 [2.45]	0.40 [2.96]	0.33 [2.49]

Table 3: Conditional sort on size and DSF

This table presents results for conditional double sorts on size and DSF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSF and short stocks with low DSF .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DSF Quintiles					DSF Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [1.64]	0.80 [2.43]	0.72 [1.99]	0.82 [2.49]	0.75 [2.25]	0.20 [2.04]	0.18 [1.78]	0.16 [1.62]	0.14 [1.37]	0.19 [1.80]	0.17 [1.58]
	(2)	0.70 [2.26]	0.74 [2.49]	0.86 [2.95]	0.75 [2.64]	0.74 [2.43]	0.04 [0.39]	0.09 [0.77]	0.11 [1.03]	0.06 [0.53]	0.11 [0.96]	0.06 [0.54]
	(3)	0.74 [2.72]	0.72 [2.70]	0.81 [3.02]	0.83 [3.20]	0.81 [2.93]	0.07 [0.64]	0.06 [0.52]	0.09 [0.81]	0.02 [0.21]	0.19 [1.75]	0.13 [1.21]
	(4)	0.82 [3.09]	0.72 [3.01]	0.70 [2.97]	0.83 [3.54]	0.90 [3.36]	0.08 [0.84]	0.07 [0.73]	0.10 [1.13]	0.05 [0.55]	0.17 [1.81]	0.12 [1.28]
	(5)	0.61 [2.76]	0.81 [4.08]	0.69 [3.53]	0.67 [3.31]	0.97 [4.39]	0.37 [3.36]	0.37 [3.32]	0.39 [3.54]	0.31 [2.85]	0.42 [3.74]	0.35 [3.15]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DSF Quintiles					DSF Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	344	343	340	341	343	36	38	38	37	38	
	(2)	108	108	108	108	108	72	74	74	74	74	
	(3)	76	77	76	77	77	131	133	134	134	133	
	(4)	67	66	67	66	67	295	303	300	308	304	
(5)	61	61	61	61	61	2517	2051	2131	2508	2902		

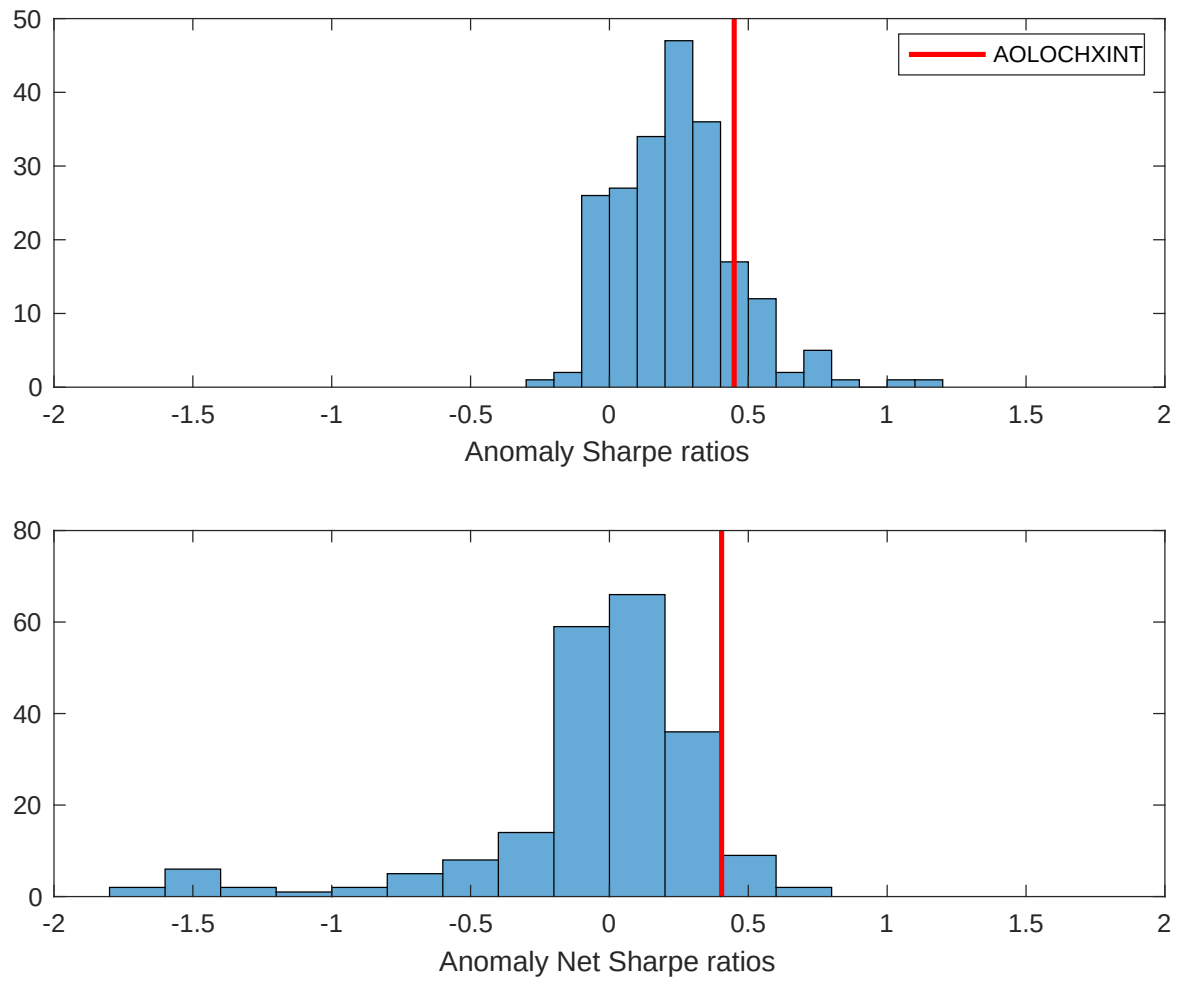


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

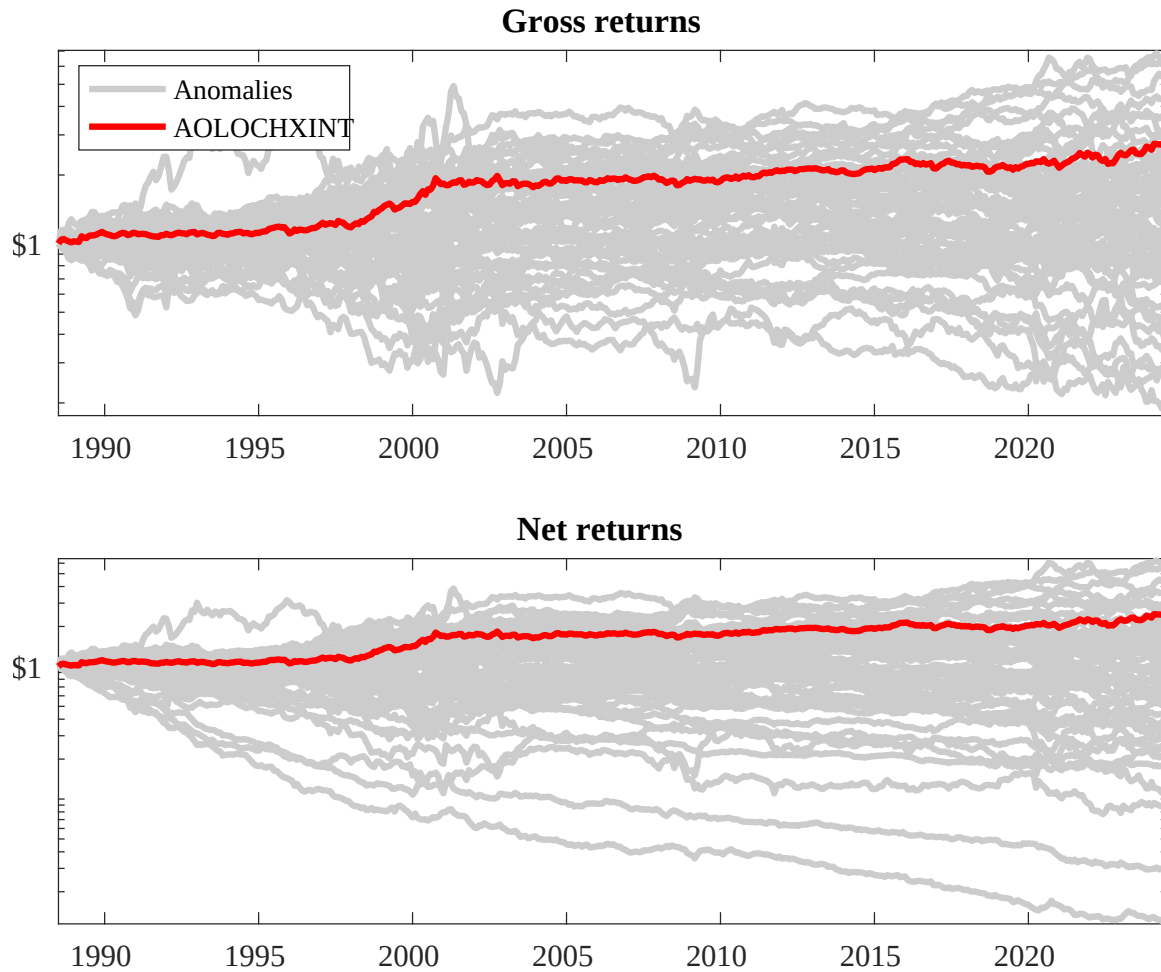
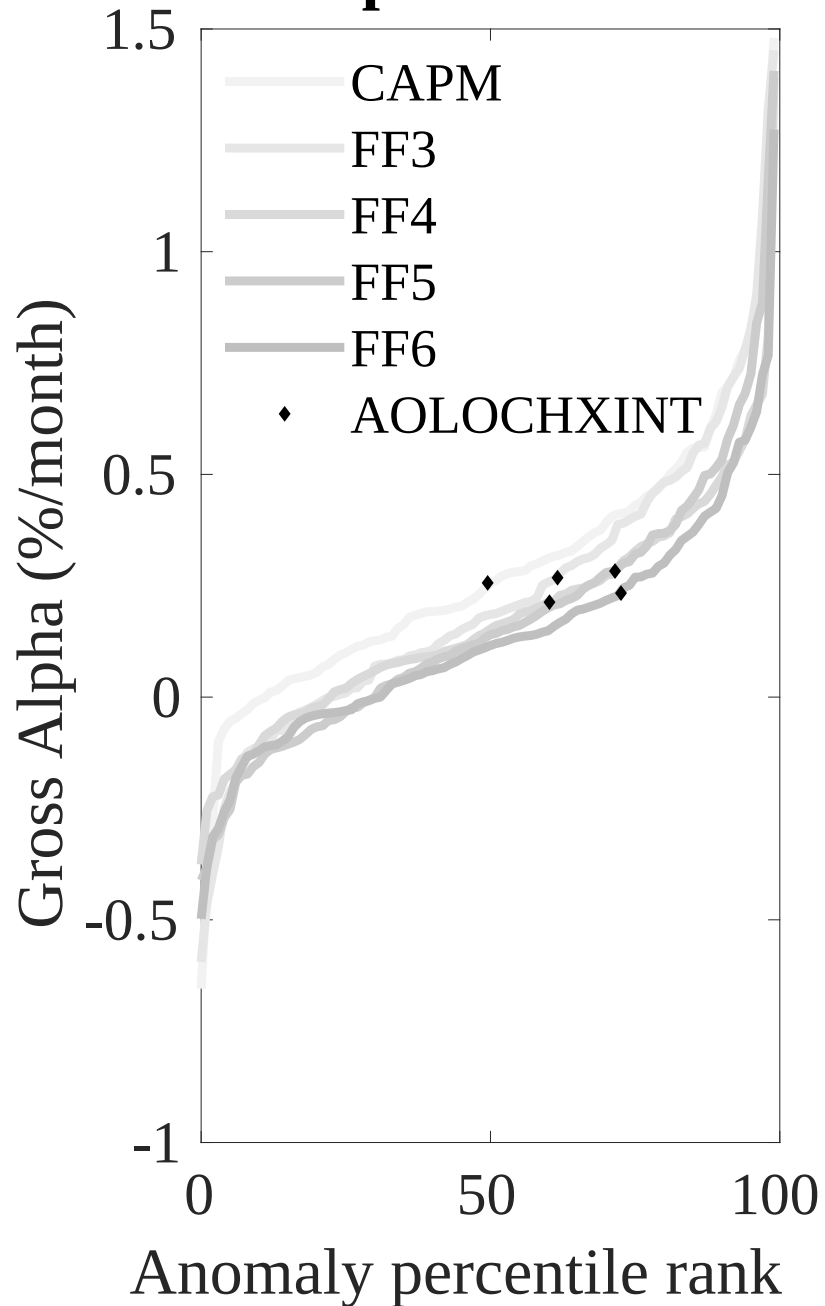


Figure 3: Dollar invested.

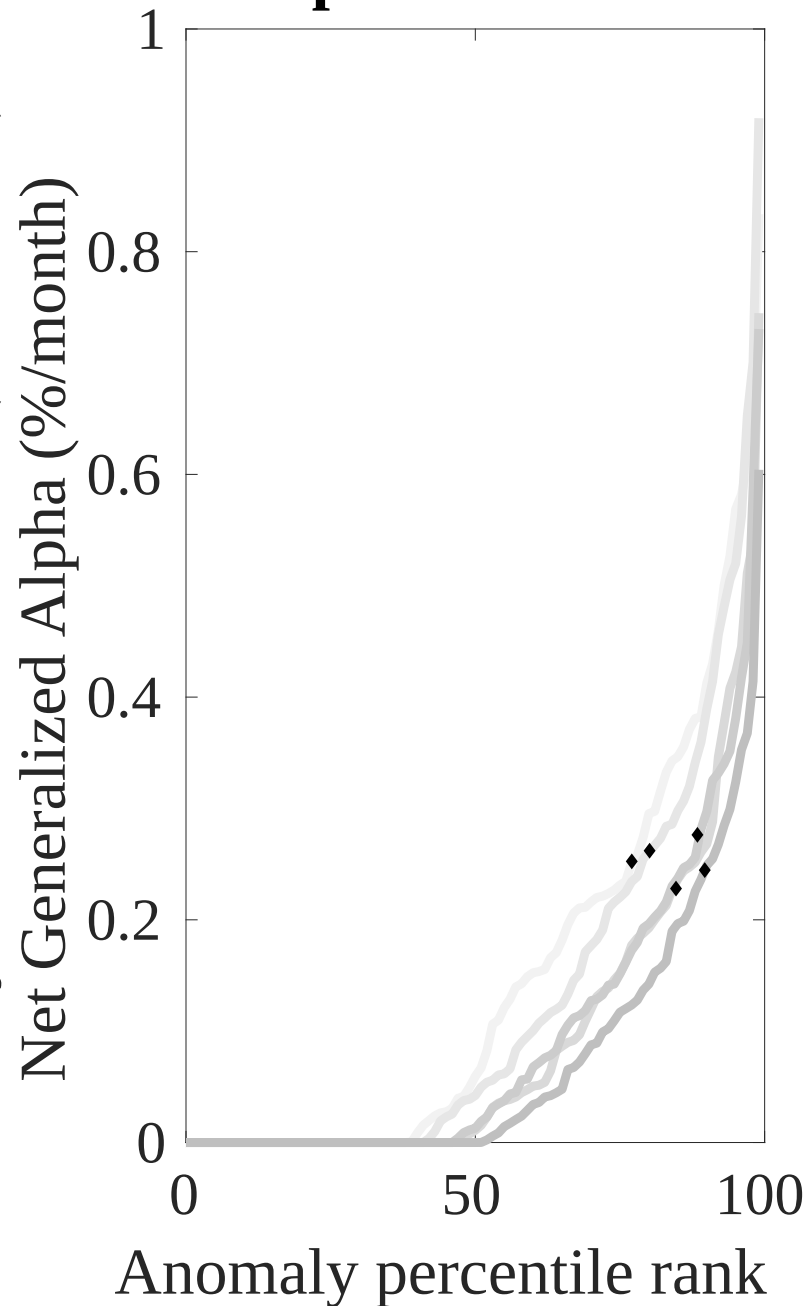
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



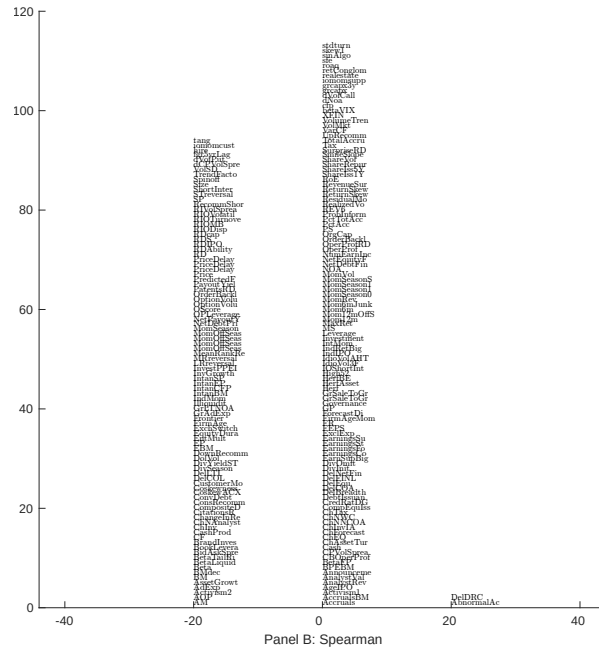
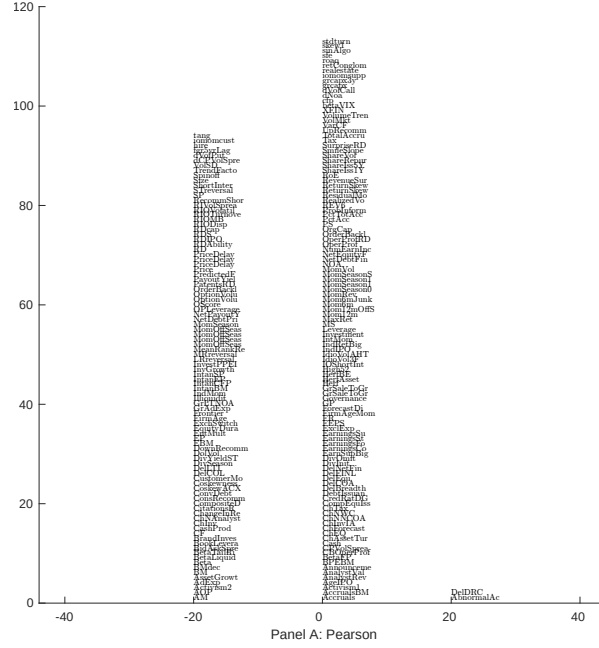


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSF. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

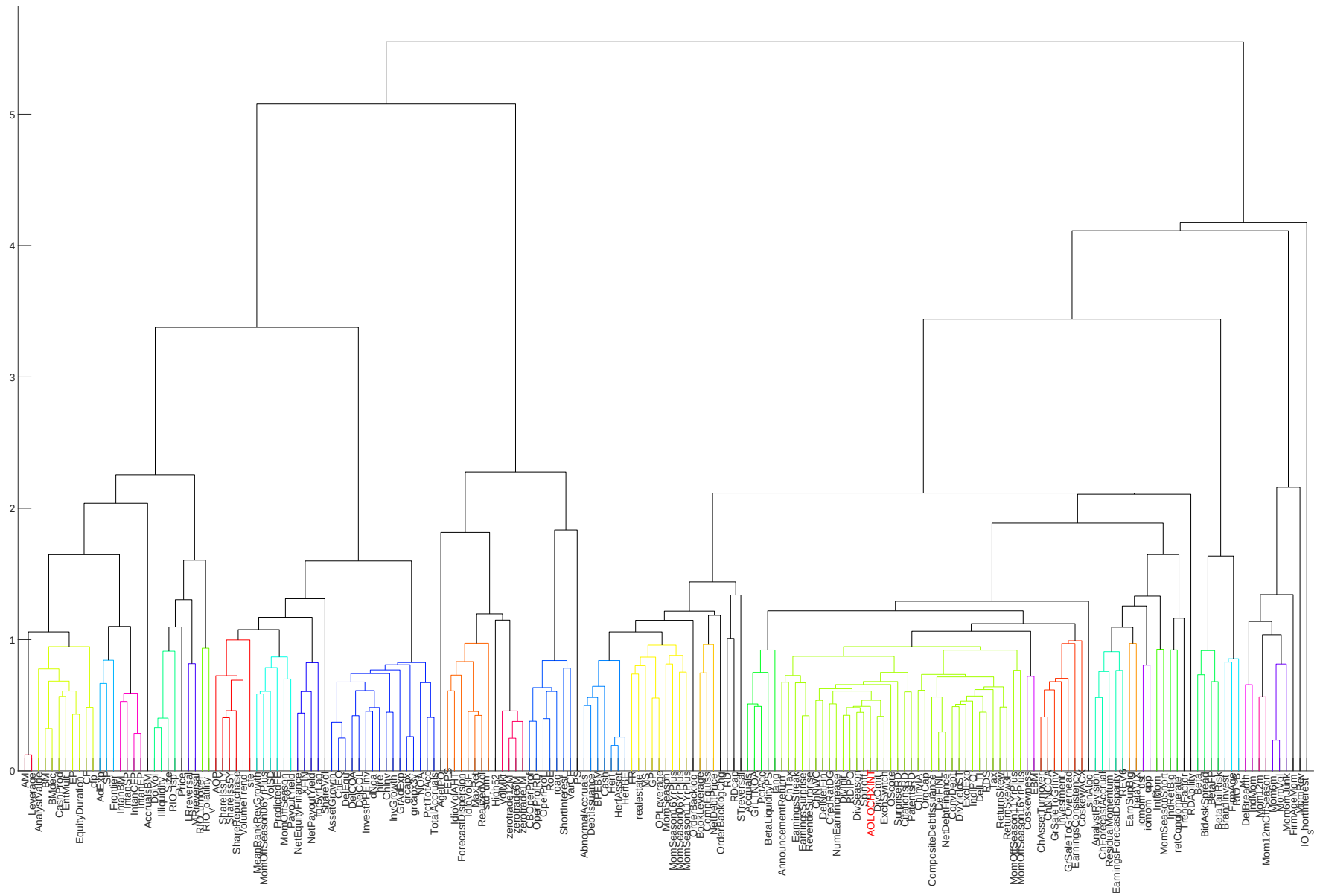


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

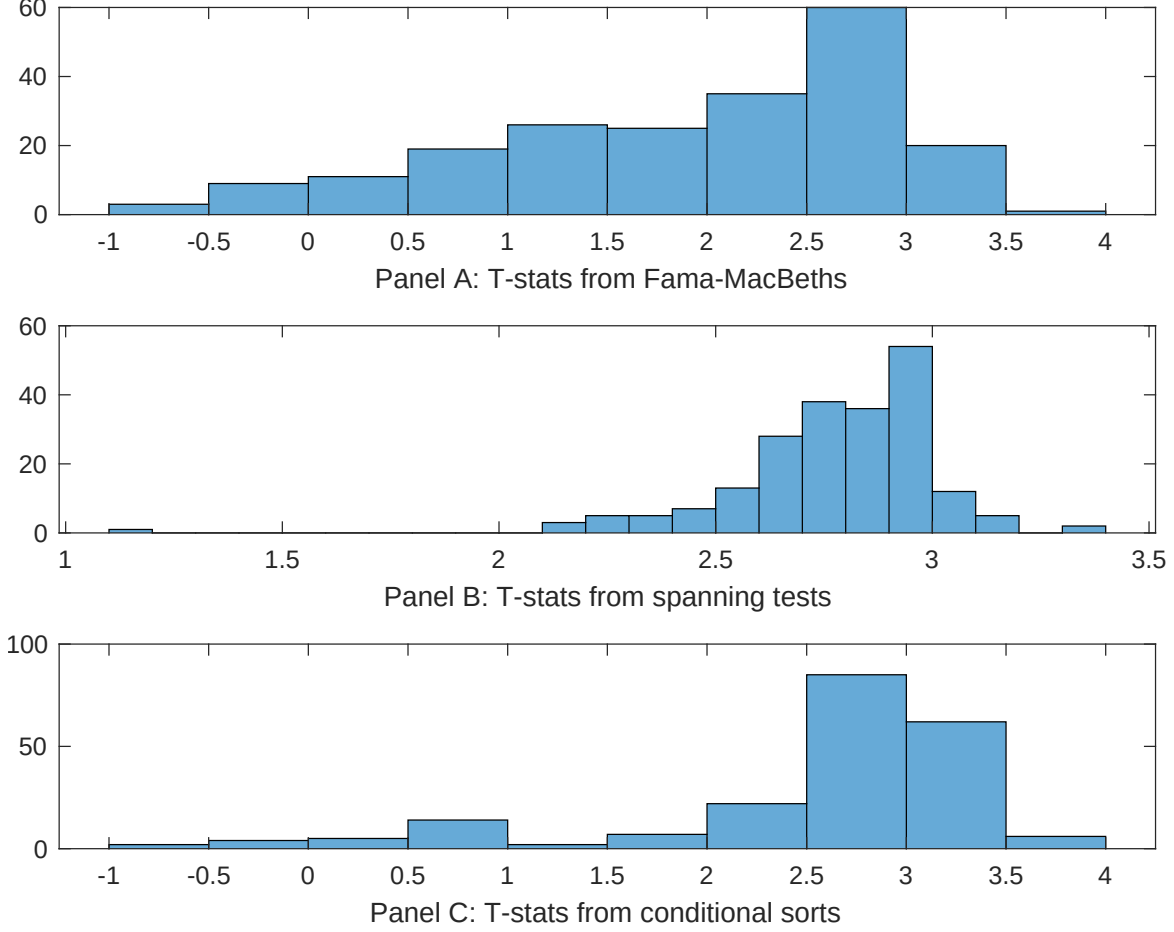


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSF. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.95]	0.66 [1.85]	0.11 [3.87]	0.11 [3.90]	0.11 [2.97]	0.11 [3.86]	0.24 [3.38]
DSF	0.35 [3.05]	0.80 [1.59]	0.25 [1.68]	0.31 [2.71]	0.36 [3.21]	0.26 [2.21]	0.28 [0.42]
Anomaly 1	0.19 [4.46]						1.00 [0.58]
Anomaly 2		0.14 [4.58]					-0.25 [-3.26]
Anomaly 3			0.44 [4.46]				0.20 [6.41]
Anomaly 4				0.14 [5.38]			0.20 [2.25]
Anomaly 5					0.45 [0.88]		-0.75 [-0.81]
Anomaly 6						0.47 [1.64]	0.67 [1.05]
# months	432	415	427	432	432	432	267
$\bar{R}^2(\%)$	0	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22 [2.54]	0.26 [2.99]	0.26 [2.93]	0.26 [3.05]	0.26 [2.96]	0.26 [3.03]	0.24 [2.71]
Anomaly 1	-12.60 [-3.09]						-8.98 [-2.02]
Anomaly 2		-1.21 [-0.96]					-1.07 [-0.85]
Anomaly 3			-0.40 [-0.15]				0.77 [0.27]
Anomaly 4				17.50 [5.03]			15.42 [4.24]
Anomaly 5					0.34 [0.10]		-0.42 [-0.12]
Anomaly 6						-7.57 [-1.87]	-1.45 [-0.33]
mkt	-1.62 [-0.75]	-1.95 [-0.89]	-2.03 [-0.92]	-3.61 [-1.68]	-1.94 [-0.86]	-1.74 [-0.80]	-3.17 [-1.41]
smb	-7.48 [-2.39]	-5.70 [-1.78]	-6.07 [-1.79]	-6.38 [-2.09]	-6.63 [-1.51]	-6.89 [-2.19]	-6.80 [-1.59]
hml	7.07 [1.65]	2.40 [0.58]	1.19 [0.30]	3.45 [0.92]	0.86 [0.22]	0.17 [0.05]	8.39 [1.76]
rmw	2.44 [0.62]	1.36 [0.34]	1.90 [0.48]	1.98 [0.51]	2.33 [0.52]	0.52 [0.13]	1.05 [0.23]
cma	-10.72 [-1.81]	-16.70 [-2.97]	-17.09 [-3.03]	-15.25 [-2.81]	-17.38 [-3.10]	-12.10 [-1.95]	-9.37 [-1.48]
umd	7.23 [3.74]	7.10 [3.40]	7.55 [2.86]	3.34 [1.60]	8.07 [2.73]	7.20 [3.66]	2.89 [0.91]
# months	432	428	428	432	432	432	428
$\bar{R}^2(\%)$	10	8	8	13	8	9	13

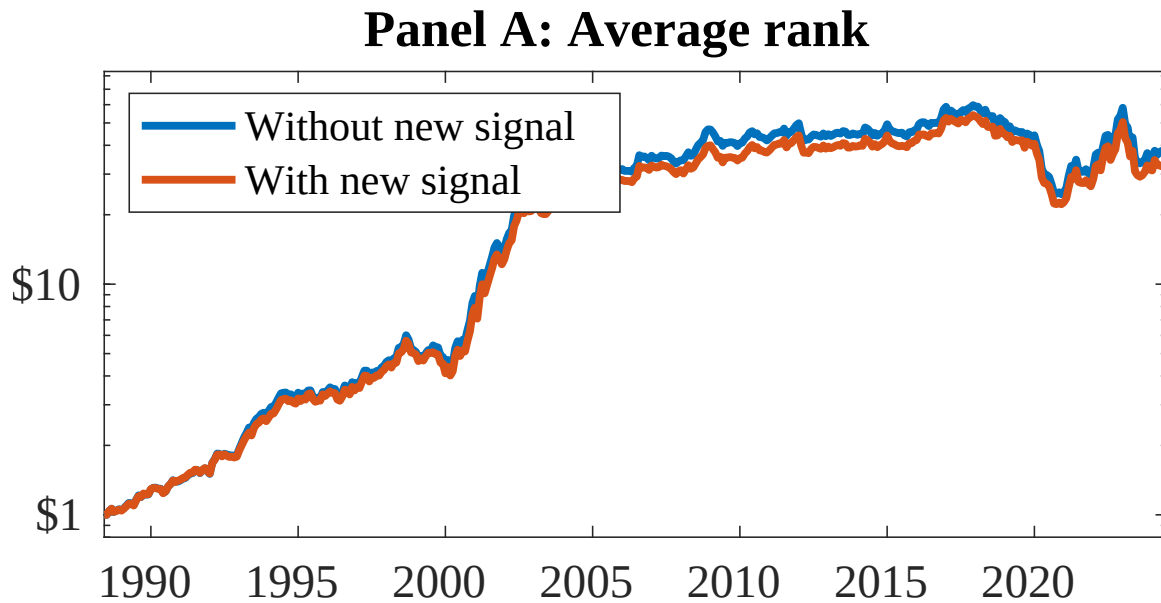


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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